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Abstract

1 Introduction

Lake ice in the Great Lakes is highly variable and has impacts ranging from economy to winter ecology. The presence and timing of ice cover influences the length of the commercial shipping season and during high ice years shipping has been entirely suspended which interrupts supply chains that rely on raw materials such as Iron Ore, Grain, etc. (Each year 135 million tons of cargo are moved in the Great Lakes which support a quarter million jobs (Martin, n.d.)). Additionally, many coastal communities rely on ice cover for winter tourism which boosts local economies. [citation] The biota of the lakes interact with ice cover which impacts nutrient loading (Nicholls, 1998), serves as a winter habitat for Diatoms (Twiss et al., 2012) and has been widely suggested to protect Lake Whitefish eggs during otherwise harsh winter conditions (Brown et al., 1993; Collingsworth et al., 2017; Lynch et al., 2015).

The timing and extent of Great Lakes ice vary considerably from year to year. For example, the long-term (1973-2025) average and standard deviation for annual maximum ice cover is 52% and 24%, respectively, which implies that peak ice cover for a given year is frequently ($\mu \pm \sigma$) as low as 28% or as high as 76%. The length of the ice season is, on average, 82 days long (based on 10% threshold) but has been as short as 23 days and as long as 148 days. This variability has significant impact on Great Lakes ecosystems and further motivates the need to understand the drivers of ice cover in order to make meaningful predictions.

Previous studies have demonstrated that, consistent with trends in air and ocean temperatures, water temperatures in the Great Lakes, are also increasing both in the surface (Austin and Colman, 2007; O'Reilly et al., 2015; Woolway and Merchant, 2018; Zhong et al., 2018) and subsurface (Anderson et al., 2021; Cannon et al., 2024). However, the rate and extent of warming in the Great Lakes varies considerably from lake to lake and even *within* each lake (Mason et al., 2016), partly due to the heterogeneity of characteristics within lakes (bathymetry, latitude, surface area) often referred to as lake morphometry. These characteristics can act as competing factors for changes in lake ice cover. For example, Sharma et al. (2019) found that mean depth influenced ice loss for lakes at similar latitudes. Richardson et al. (2024) hypothesized that, globally, lake ice phenology will respond non-linearly to climate-change such that lakes in geographically colder (warmer) regions experience increased (decreased) interannual variability; furthermore, if ice duration becomes too short, variability may decrease. This is mostly consistent with the findings of Lin et al. (2022), who showed that regionally the Great Lakes have been experiencing increased variability since 1998.

Assel (2005) classified ice seasons as either mild, typical or severe and found that. . .

Numerical ice forecasts are high resolution and go beyond the sub-regional scale but they are computationally expensive and have showed varied skill with some lakes and years more prone to error (Anderson

et al., 2018). Subregional ice forecasts Venumuddula et al., 2024 have proved advantageous over forecasts at larger scales...

To further understanding of the physical drivers of ice cover formation and decay, we analyze 54 years of daily ice cover in the Great Lakes at sub-lake scales. By evaluating ice cover aggregated into subregions, we gain insight into changes and relationships occurring at the sub-lake scale that might otherwise be masked by those at the lake-wide scale. Furthermore, the heterogeneity of physical characteristics (e.g. depth) within the Lakes motivates an analysis of how those varying characteristics may be impacting long-term changes in ice cover. Additionally, we leverage *in-situ* observational data such as buoys and coastal meteorological station data to assess how locally-specific atmospheric and lake conditions impact the evolution ice cover over the course of a given ice season.

2 Methods

2.1 Overview

Following the work of Mason et al. (2016), we aggregate gridded ice cover data into 22 regions defined by the Great Lakes Aquatic Habitat Framework (L. Wang et al., 2015) (hereafter “subregions”, see figure 1) which have been thoughtfully defined to delineate the Great Lakes (and watersheds) by various factors including circulation patterns, thermal stratification, as well as physiochemical and biological characteristics (see L. Wang et al., 2015 for more details.) Once aggregated, ice metrics are calculated for each of the 54 ice seasons on record: annual maximum ice cover, seasonal average ice cover and ice phenology (duration, onset, breakup). Though these ice metrics are closely correlated, each provides unique information about a given ice season and show different relationships with morphometrics as can be seen in the results section.

The first component of analysis is identifying what statistically significant ice trends, if any, exist at these sub-lake scales, and how those changes might relate to the characteristics (e.g. morphometrics) of a given subregion. The second component addresses how atmospheric phenomena and the thermodynamic state of a lake or subregion play a role in dictating ice cover on the local scale. This is accomplished by using *in-situ* observations specific to each subregion to identify correlations between atmosphere and lake conditions throughout the year and ice phenology.

While much previous work has identified the significant influence of teleconnections on (Great) Lakes ice cover (Bai et al., 2012; Richardson et al., 2024; J. Wang et al., 2018), our focus is on subregional ice formation (i.e. differences between . . .

By identifying subregional ice trends, we gain deeper insight into how lake ice is changing and what might be expected in the future at a more detailed level for each specific region. This specificity is valuable when considering nearly all of the motivations for studying lake ice: commercial navigation occurs along well-defined shipping routes, residents interact with ice along a specific shoreline and ecosystems exists as microcosms in the Great Lakes [is this true?]. Additionally, relating changes in ice cover back to physical characteristics provides clues as to what drives the change and what makes a specific subregion or lake more susceptible to that change. These results can have implications for large lakes globally especially those at similar latitudes or with similar physical characteristics.

First, ice metrics for each of the subregions is calculated and statistically significant trends are tested for. Next, relationships between ice trends and lake characteristics (morphometrics) are identified. Finally, meteorological and limnological data are leveraged to identify how the timing and magnitude of regional phenomena impact ice evolution. This analysis is described in detail in the following subsections.

2.2 Metrics

We define an ice season as the period between November and the following May and, hereafter, refer to a given season by the calendar year in which January occurs (e.g. the 1996 - 1997 winter is referred to as ice year 1997.) The following ice metrics are used to analyze ice cover: annual maximum ice cover (AMIC), seasonal average ice cover (JFM), ice duration, ice onset and ice breakup. Annual max is simply the highest daily value reported in a given ice season and is expressed as percentage. The seasonal average is defined as the mean ice cover from January 1 through March 30. We restrict the average to January, February and March because ice data was not consistently reported outside this window during the 1970's and 1980's. (If alternatively, one averages over the entire season, systematic error can be introduced by implicitly treating missing data as zero. This is particularly critical for doing trend analysis and we recommend future work is cautious of this issue.)

In addition to expressing ice cover as a percentage, we can also consider the timing and duration of ice cover in a given subregion. We define the ice duration (??) as the total number of (non-consecutive) days in which ice was above the threshold of 10%. Using the same threshold, we define ice onset (ice breakup) as the first (last) day in a season during which ice cover was observed. Although this includes data outside the JFM date range, the aforementioned systematic error is most likely avoided since ice cover reporting generally began well below 10% and persisted until it was lower. [I think maybe David found this to be untrue on the local scale?, if so we could increase the threshold]. Last, we can also consider the date when AMIC occurred.

$$\text{Ice Duration} = \sum_{d=t_0}^{t_f} \begin{cases} 1 & i(d) \geq 10 \\ 0 & i(d) < 10 \end{cases} \quad (1)$$

where $i(d)$ is ice cover on day, d which ranges between Nov 1 - May 31.

2.3 Ice Trends and Morphometrics

To assess changes in ice cover on the local scale, the above ice metrics are first calculated for each of the 22 subregions and each of the 53 ice seasons. These metrics can be evaluated visually as time series which is directly useful for understanding how ice varies in given regions. Next, linear regression is applied to each ice metric which provides a more quantitative evaluation of changes in ice phenology. We apply ~~ordinary least squares and~~ Theil-Sen (SIEGEL, 1982)[and also STL?] to derive the linear trend as well as applying Mann-Kendall (cite) to test for statistical significance. (If a linear trend is not deemed significant at the 95% level or if confidence intervals include a trend of 0, we exclude that metric/subregion trend from further analysis.)

Following the trend assessment, we identify correlations between the statistically significant trends and the physical characteristics (e.g. morphometrics) of each subregion. The variables assessed include mean [also do max?] depth, surface area, latitude and longitude. We hypothesize that subregions with similar characteristics will also show similar changes in ice cover (e.g. deeper regions will respond differently than shallower regions to the observed surface air temperature increase over time).

2.4 Correlations with *in-situ* data

2.5 Data

High-resolution ice cover in the Great Lakes has been recorded since 1973¹ coinciding with the advent of consistent satellite overpass in the region. This data set is made possible by the North American Ice Services (NAIS: comprised of US National Ice Center and Canadian Ice Service) who generate daily (semi-weekly prior

¹Though some ice measurements pre-date 1973, they did not include comprehensive coverage of the lake surface so will not be considered in this work [(see CITE: TM-121 for details.)]

to 2011) data using a variety of sources including remote-sensing data (synthetic-aperture radar, infrared and visible imagery), ship reports, ice reconnaissance flights and analyst knowledge. This data set was standardized by Yang et al. (2020) via re-sampling onto a 1.25 KM grid for the entire period of record and also temporally interpolated from semi-weekly to daily frequency. We use this standardized data set (hereafter: NAIS data) for all of our analysis.

Meteorological data is available from the Great Lakes

3 tasks

- change point analysis for each region?
 - does a stat sig. chng pt negate a stat sig trend?
- should we subset recent data to match frequency of older data? or should we repeate analysis using the daily interpolated data?
- include model data (or mooring data) for subregions stratification or heat content? this is a biiig can of worms
- mooring data might be doable but limited regions and years
- calculate first last and ndays using the ACTUAL eqn you show
- also calculate the DAY OF MAX
- Theil-Sen estimates
- read theil-sen paper (siegel 1982)
- confidence intervals (how to show on plot?)
- is some form of trend analysis possible using a subset of the data?
 - specifically, if we categorize high/med/low ice years can we perform linear regresssion on each set?
 - seems like discontinuity (i.e. varying delta t) would be a problem

4 Results

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